

Report

Title: Configuring Apache Web Server on AWS EC2

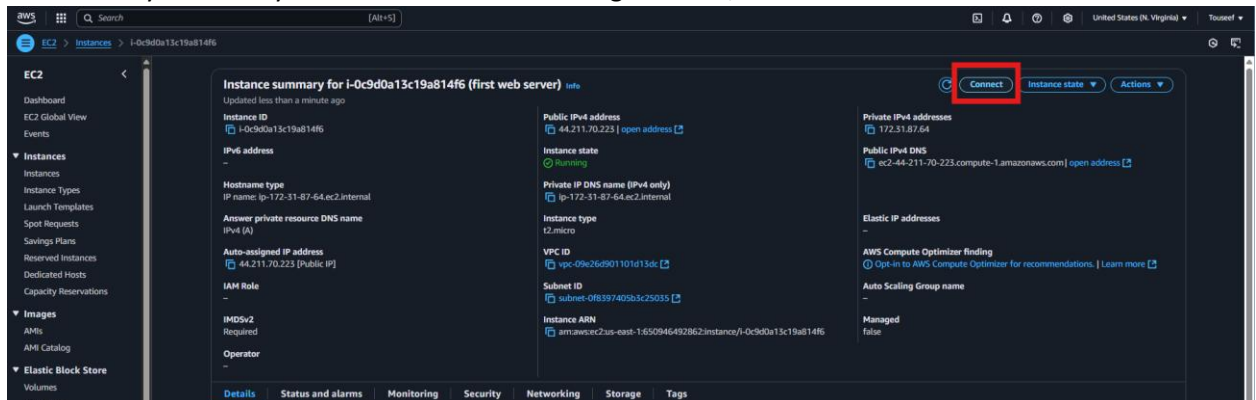
Setup

Create a fresh EC2 instance with following configs;

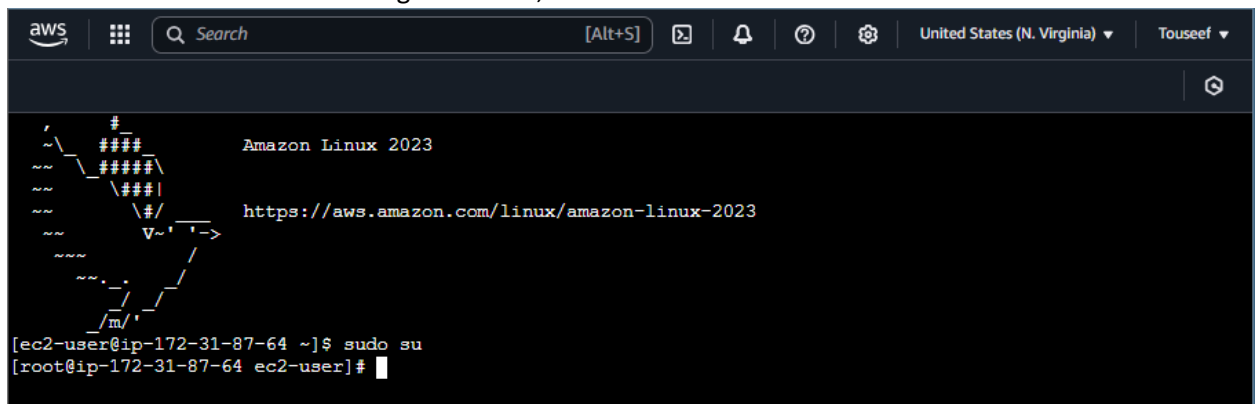
- No Key Value Pair Encryption
- Allow http/https Traffic
- Everything else to default values

Procedure

1. Connect to your freshly created AWS instance using browser;



2. Switch to root user with following command;



- Update the machine packages with following command;

```
aws [Search] [Alt+S] [United States (N. Virginia)] Touseef
[root@ip-172-31-87-64 ec2-user]#
[root@ip-172-31-87-64 ec2-user]#
[root@ip-172-31-87-64 ec2-user]#
[root@ip-172-31-87-64 ec2-user]# yum update -y
Amazon Linux 2023 Kernel Livepatch repository 112 kB/s | 14 kB 00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-87-64 ec2-user]#
```

- Install the Apache web server using the following command;

```
[root@ip-172-31-87-64 ec2-user]#
[root@ip-172-31-87-64 ec2-user]#
[root@ip-172-31-87-64 ec2-user]# yum install -y httpd
Last metadata expiration check: 0:00:44 ago on Sat Mar 15 04:49:59 2025.
Dependencies resolved.
```

Package	Architecture	Version	Repository	Size
Installing:				
httpd	x86_64	2.4.62-1.amzn2023	amazonlinux	48 k
Installing dependencies:				
apr	x86_64	1.7.5-1.amzn2023.0.4	amazonlinux	129 k
apr-util	x86_64	1.6.3-1.amzn2023.0.1	amazonlinux	98 k
generic-logos-httpd	noarch	18.0.0-12.amzn2023.0.3	amazonlinux	19 k
httpd-core	x86_64	2.4.62-1.amzn2023	amazonlinux	1.4 M
httpd-filesystem	noarch	2.4.62-1.amzn2023	amazonlinux	14 k
httpd-tools	x86_64	2.4.62-1.amzn2023	amazonlinux	81 k
libbrotli	x86_64	1.0.9-4.amzn2023.0.2	amazonlinux	315 k
mailcap	noarch	2.1.49-3.amzn2023.0.3	amazonlinux	33 k
Installing weak dependencies:				
apr-util-openssl	x86_64	1.6.3-1.amzn2023.0.1	amazonlinux	17 k
mod_http2	x86_64	2.0.27-1.amzn2023.0.3	amazonlinux	166 k
mod_lua	x86_64	2.4.62-1.amzn2023	amazonlinux	61 k

```
Transaction Summary
Install 12 Packages

Total download size: 2.3 M
Installed size: 6.9 M
Downloading Packages:
(1/12): apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64.rpm 437 kB/s | 17 kB 00:00
(2/12): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm 2.7 MB/s | 129 kB 00:00
(3/12): apr-util-1.6.3-1.amzn2023.0.1.x86_64.rpm 1.9 MB/s | 98 kB 00:00
(4/12): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm 1.0 MB/s | 19 kB 00:00
(5/12): httpd-2.4.62-1.amzn2023.x86_64.rpm 2.6 MB/s | 48 kB 00:00
(6/12): httpd-filesystem-2.4.62-1.amzn2023.noarch.rpm 735 kB/s | 14 kB 00:00
(7/12): httpd-core-2.4.62-1.amzn2023.x86_64.rpm 29 MB/s | 1.4 MB 00:00
(8/12): httpd-tools-2.4.62-1.amzn2023.x86_64.rpm 2.1 MB/s | 81 kB 00:00
(9/12): libbrotli-1.0.9-4.amzn2023.0.2.x86_64.rpm 9.3 MB/s | 315 kB 00:00
(10/12): mailcap-2.1.49-3.amzn2023.0.3.noarch.rpm 1.6 MB/s | 33 kB 00:00
(11/12): mod_http2-2.0.27-1.amzn2023.0.3.x86_64.rpm 6.5 MB/s | 166 kB 00:00
(12/12): mod_lua-2.4.62-1.amzn2023.x86_64.rpm 2.6 MB/s | 61 kB 00:00
```

5. Start the installed Apache web server using the following command;

```
aws | [Search] [Alt+S] | United States (N. Virginia) | Touseef |  
[root@ip-172-31-87-64 ec2-user]#  
[root@ip-172-31-87-64 ec2-user]#  
[root@ip-172-31-87-64 ec2-user]# systemctl start httpd.service  
[root@ip-172-31-87-64 ec2-user]#  
[root@ip-172-31-87-64 ec2-user]#
```

6. Also enable the installed Apache web server service so that it can auto start when the EC2 instance restarts. Use the following command;

```
aws | [Search] [Alt+S] | United States (N. Virginia) | Touseef |  
[root@ip-172-31-87-64 ec2-user]#  
[root@ip-172-31-87-64 ec2-user]#  
[root@ip-172-31-87-64 ec2-user]# systemctl enable httpd.service  
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.  
[root@ip-172-31-87-64 ec2-user]#
```

7. Create a custom index.html file at /var/www/html/index.html path of your machine which would be automatically returned by the web server upon client web request.

```
aws | [Search] [Alt+S] | United States (N. Virginia) | Touseef |  
[root@ip-172-31-87-64 ec2-user]#  
[root@ip-172-31-87-64 ec2-user]#  
[root@ip-172-31-87-64 ec2-user]#  
[root@ip-172-31-87-64 ec2-user]# echo "My Web Server is Live!" > /var/www/html/index.html  
[root@ip-172-31-87-64 ec2-user]#  
[root@ip-172-31-87-64 ec2-user]#
```

8. Now copy the public IP of your EC2 machine and send the http request from your browser. You should be able to see content of your index.html file in the browser body.

